Practical-1

Problem Definition:

Design a 4-input arithmetic-control logic function H(A,B,C,D) = 1 if the 4-bit input (A B C D)

matches any of six specific patterns(select by yourself; otherwise H =
0).

You must:

1. Choose patterns.

2. Derive the Sum-of-Products form (explicit minterms).

3. Attempt simplification with a Karnaugh map and with Boolean algebra; explain differences.

4. Decide which implementation style (NAND-only, NOR- only, or mixed) is most suitable and explain why.

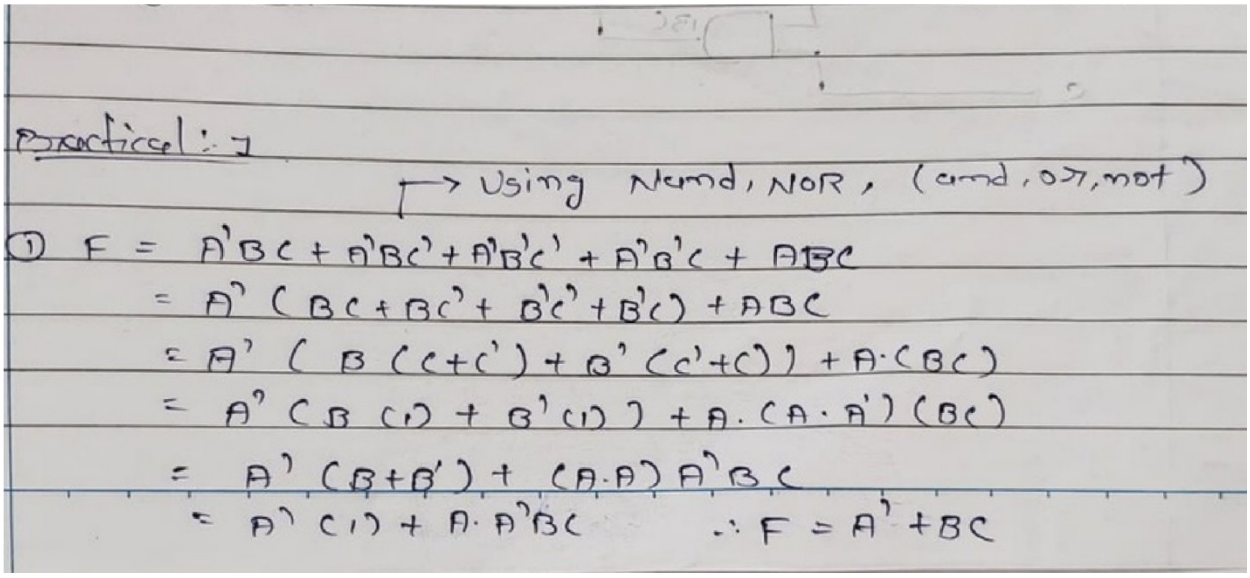
5. Implement the final design in TinkerCad and demonstrate operation with test vectors.

6. Convert the current SOP to a minimal Product-of-Sums (POS) and compare NOR-only implementation cost with NAND-only.

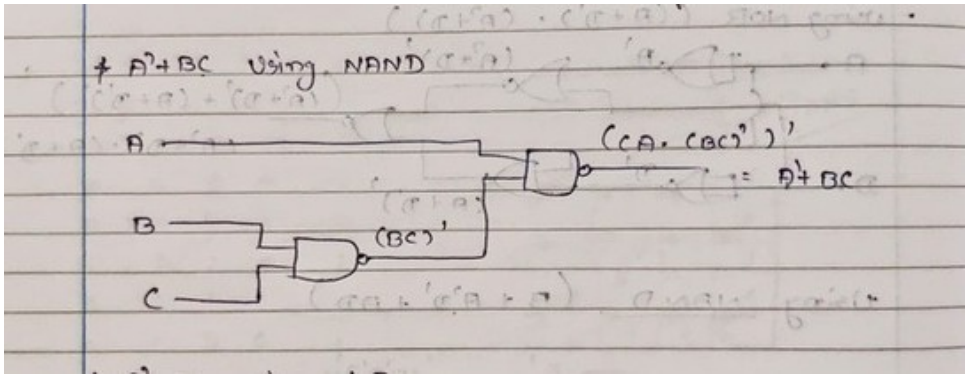
Tools/Technology To Be Used:

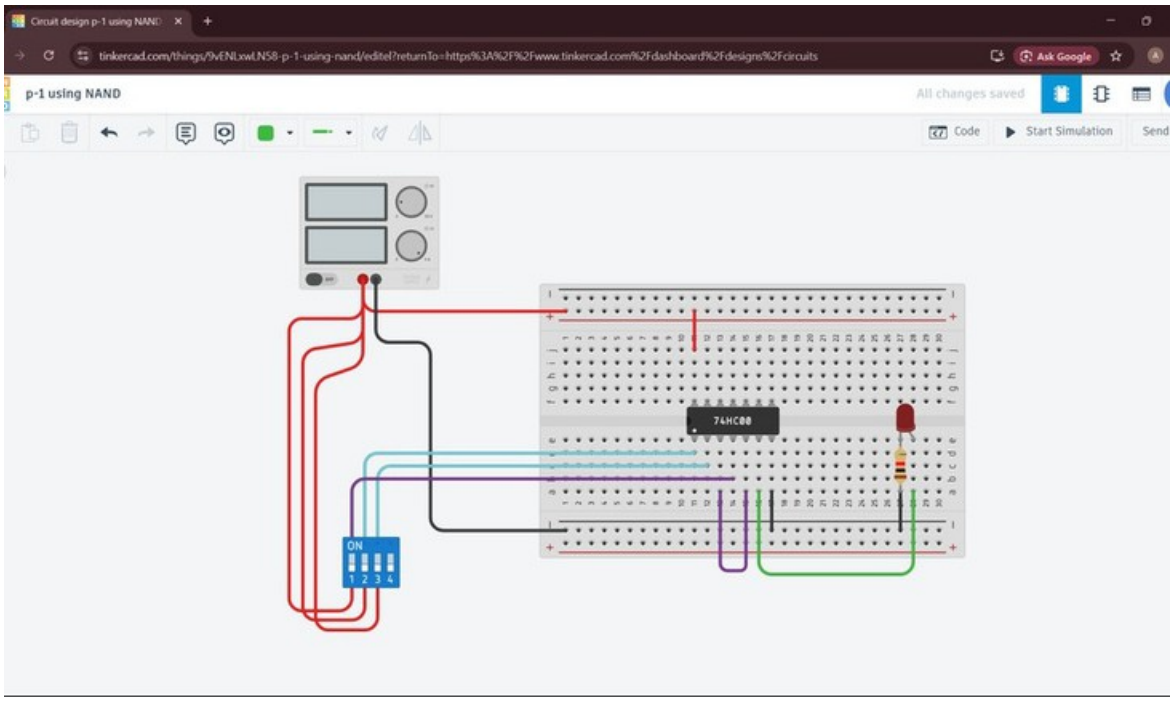
- Logic design: pen and paper
- Gate simulation & wiring: TinkerCad Circuits
- Optional: Multi-input gate ICs or build using cascaded 2-input gates.

1) simplification with Boolean algebra:

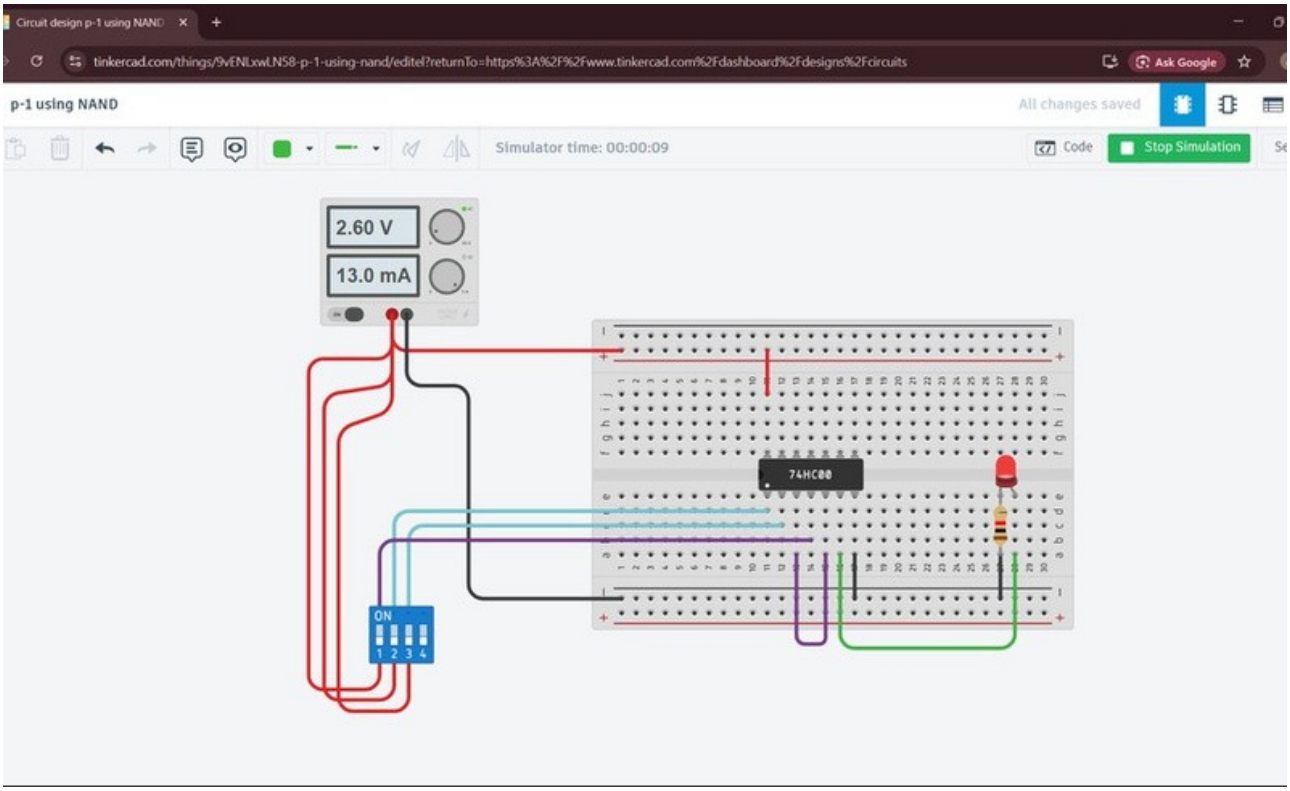


1) Using Nand gate:

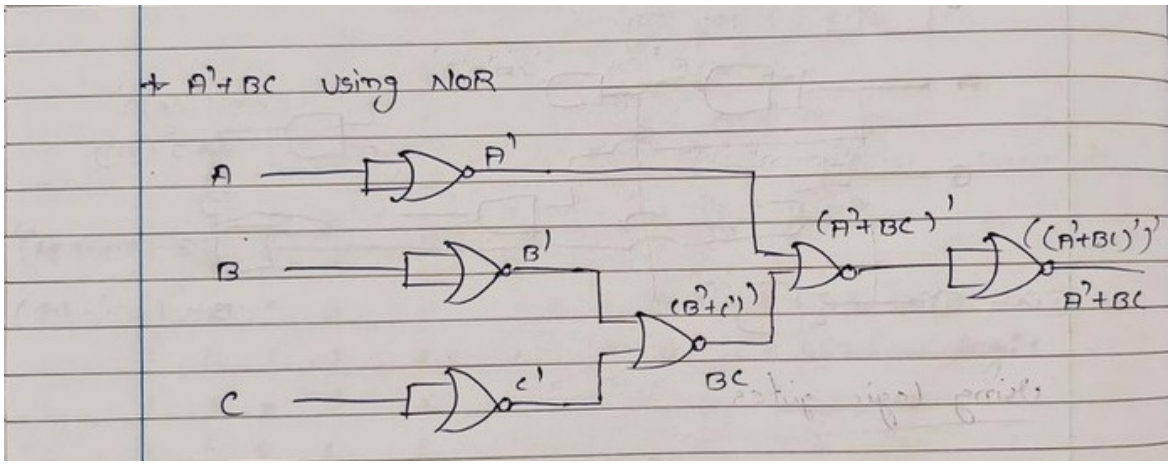


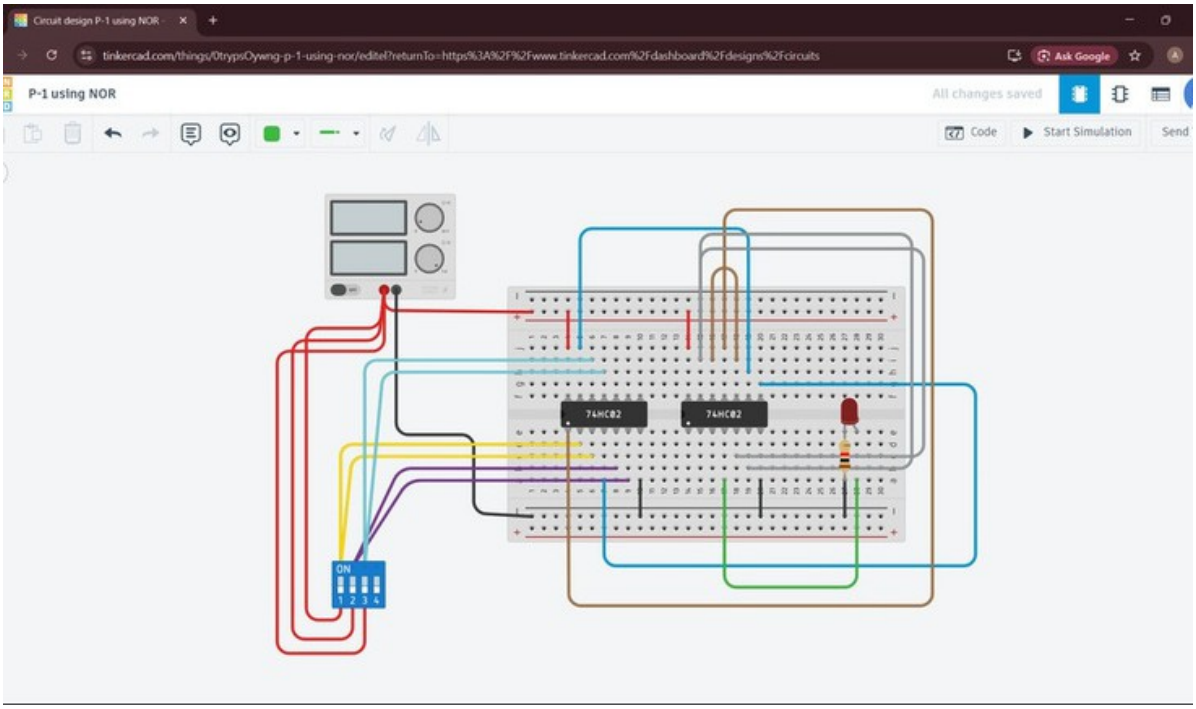


Output:

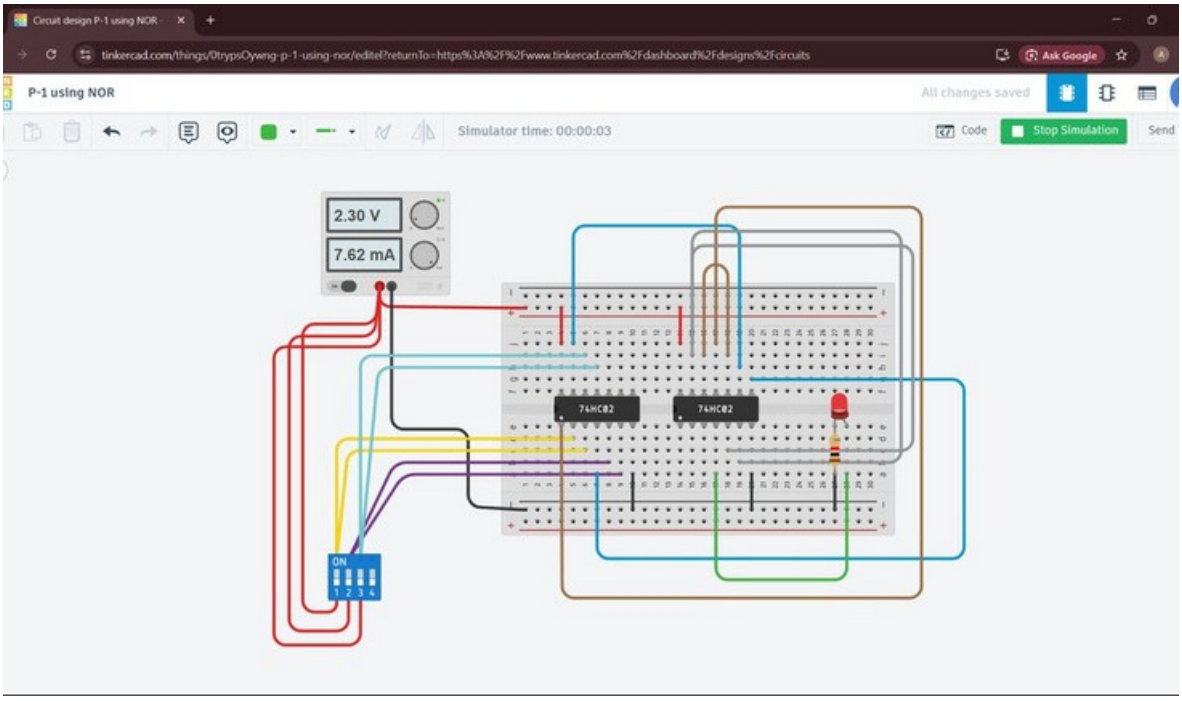


2) Using Nor gate:

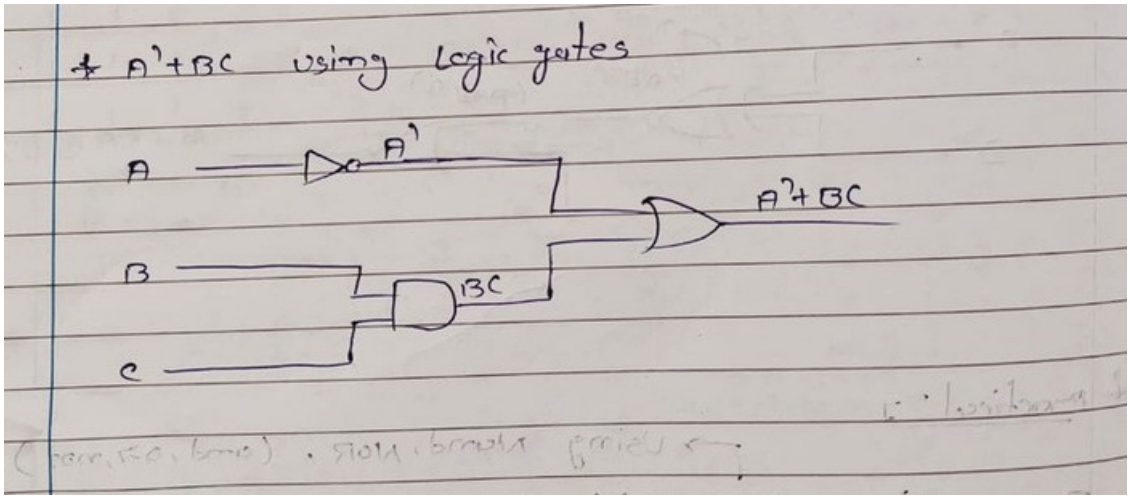


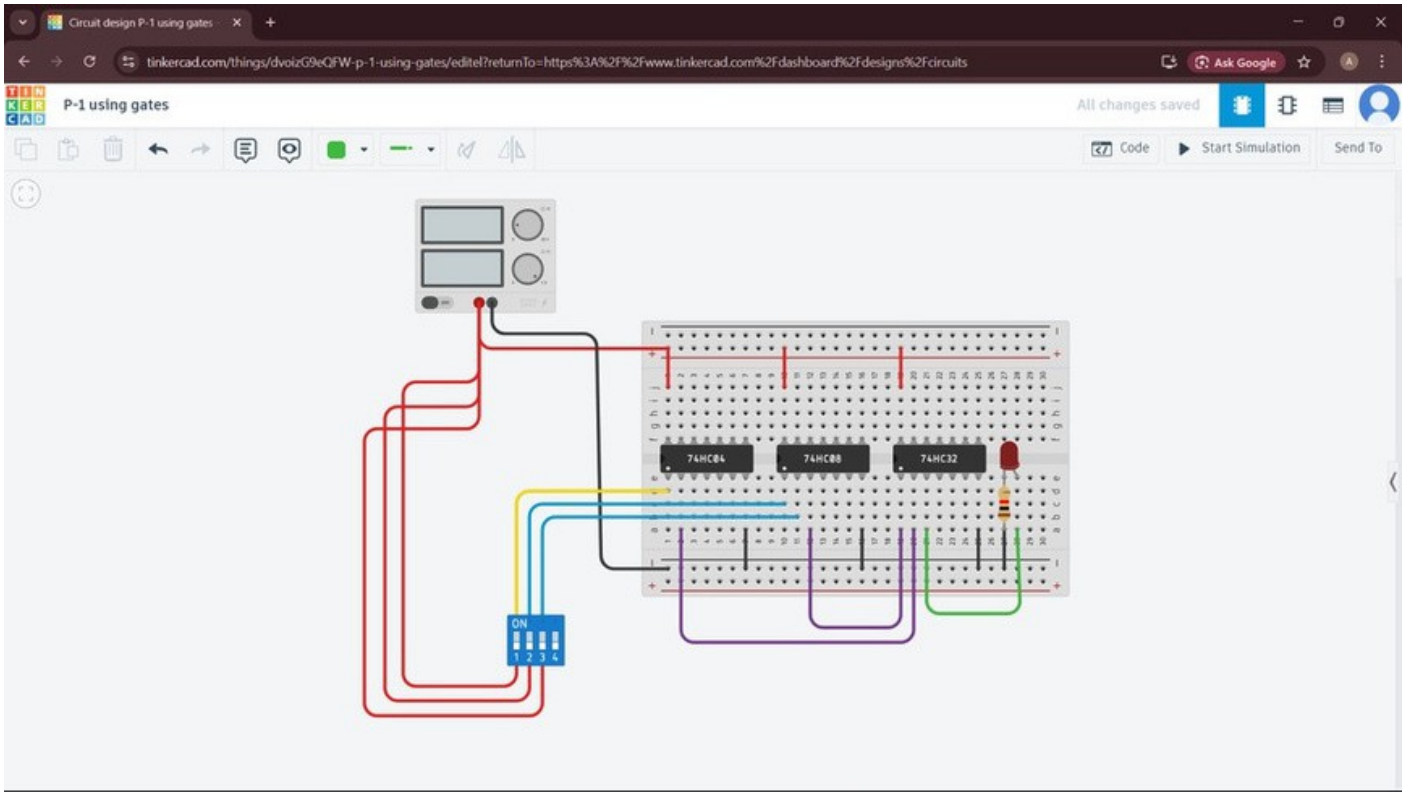

Circuit design P-1 using NOR
tinkercad.com/things/0trypsOywng-p-1-using-nor/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard%2Fdesigns%2Fcircuits
Ask Google
P-1 using NOR
All changes saved
Code
Start Simulation
Send
74HC02
74HC02
ON
1 2 3 4

Output:

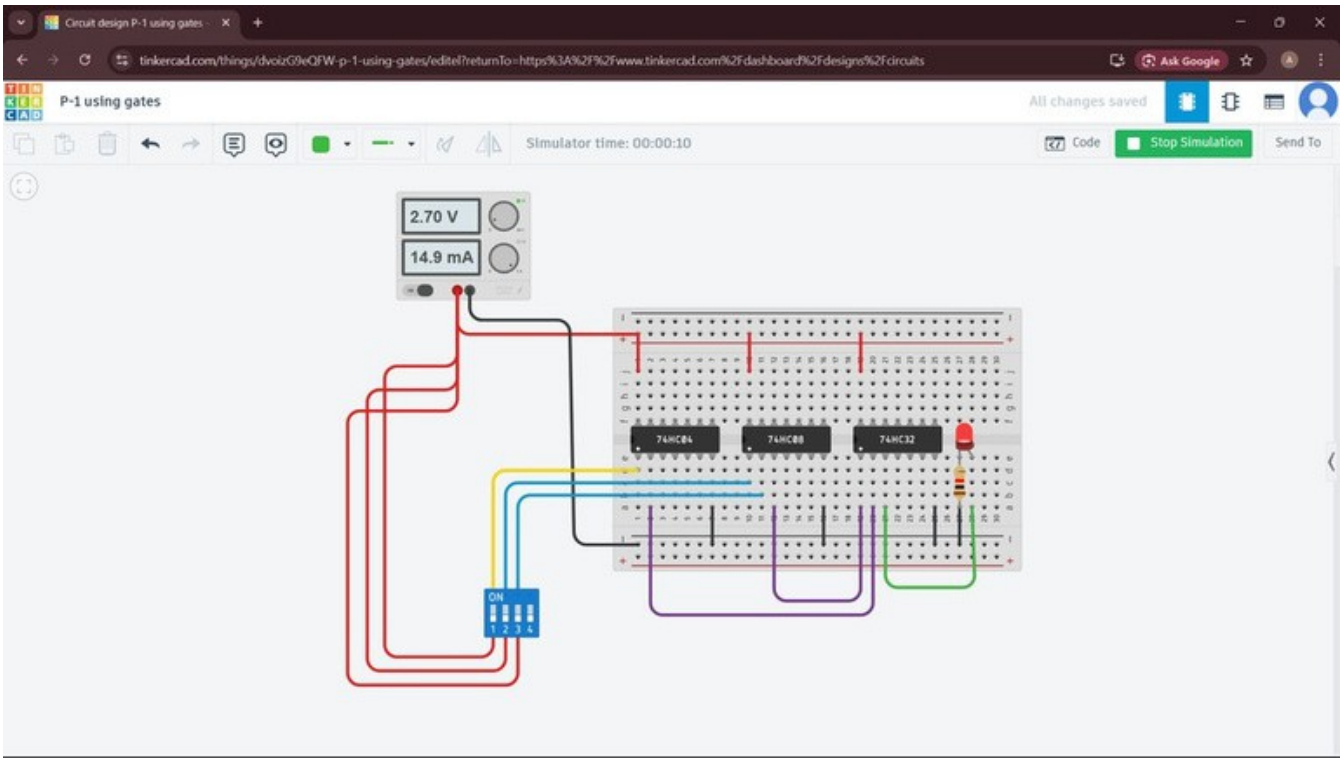

Circuit design P-1 using NOR
tinkercad.com/things/0trypsOywng-p-1-using-nor/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard%2Fdesigns%2Fcircuits
Ask Google
P-1 using NOR
All changes saved
Simulator time: 00:00:03
Code
Stop Simulation
Send
2.30 V
7.62 mA
74HC02
74HC02
ON
1 2 3 4

3) Using And ,Or, Not gate:

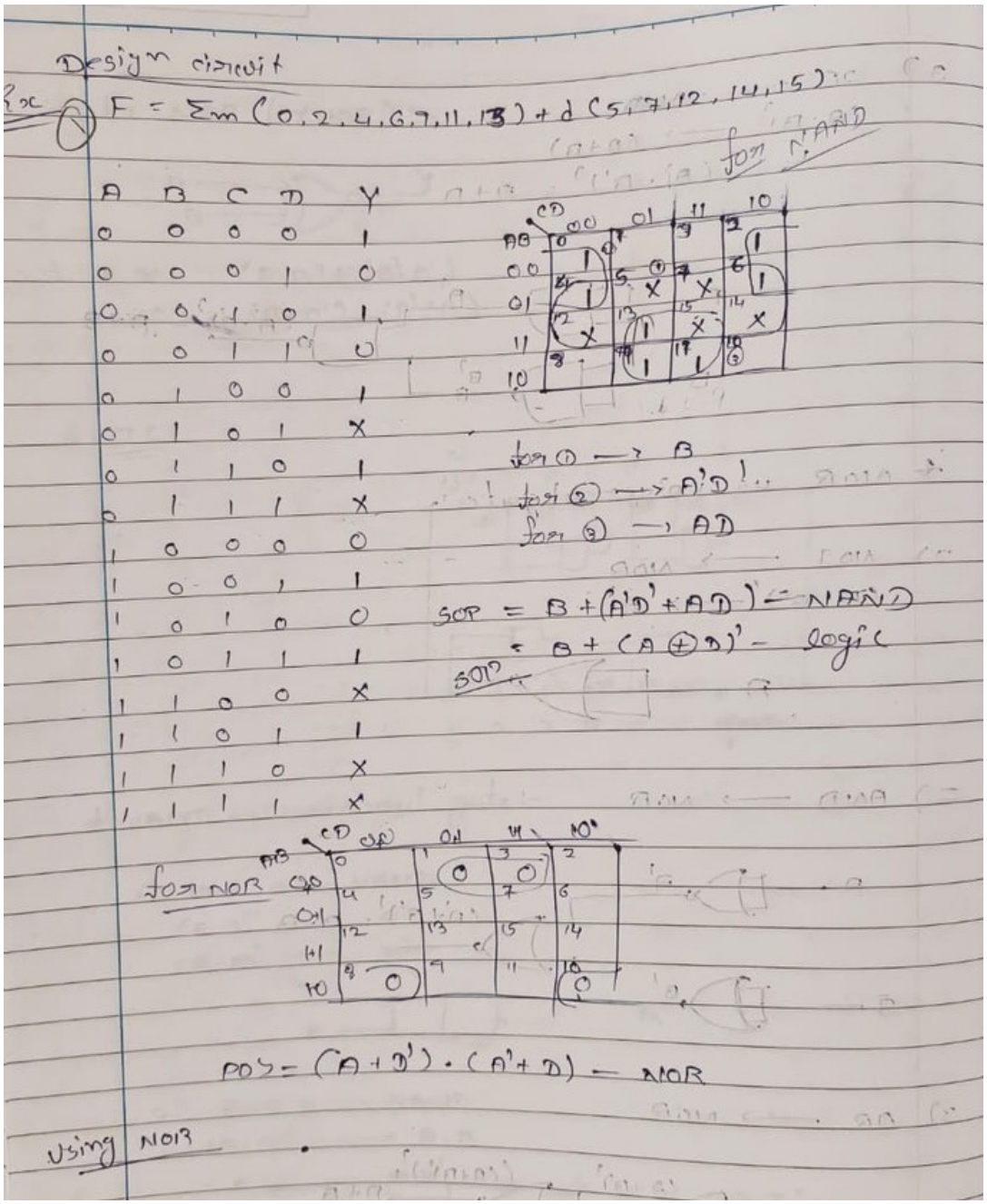




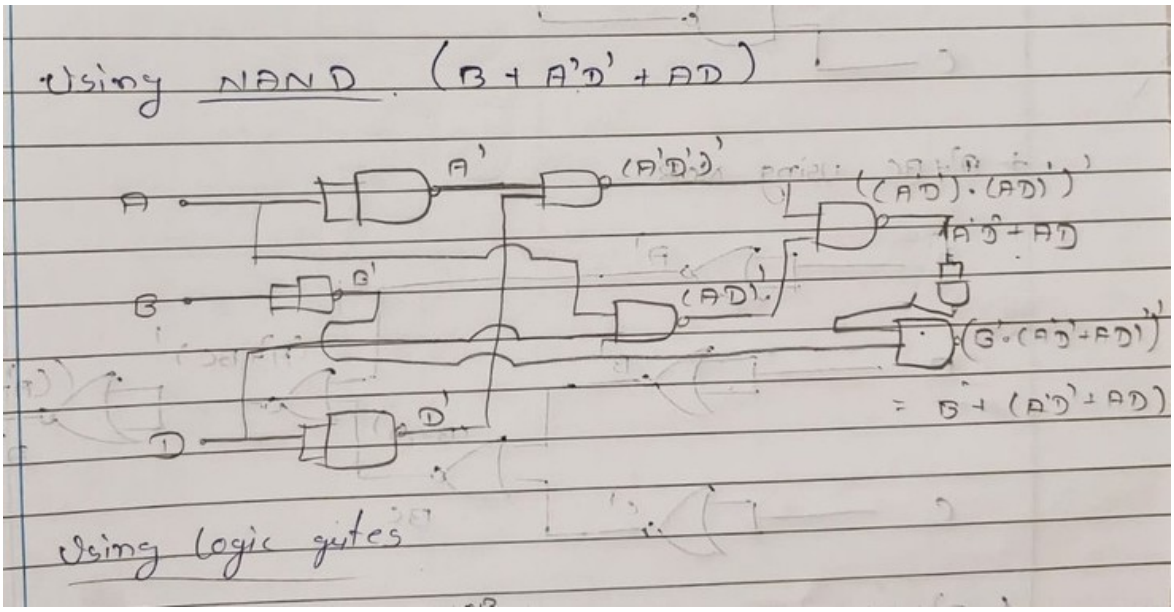
Output:


P-1 using gates
All changes saved
Simulator time: 00:00:10
Code
Stop Simulation
Send To
2.70 V
14.9 mA
74HC04
74HC08
74HC32
ON
1 2 3 4

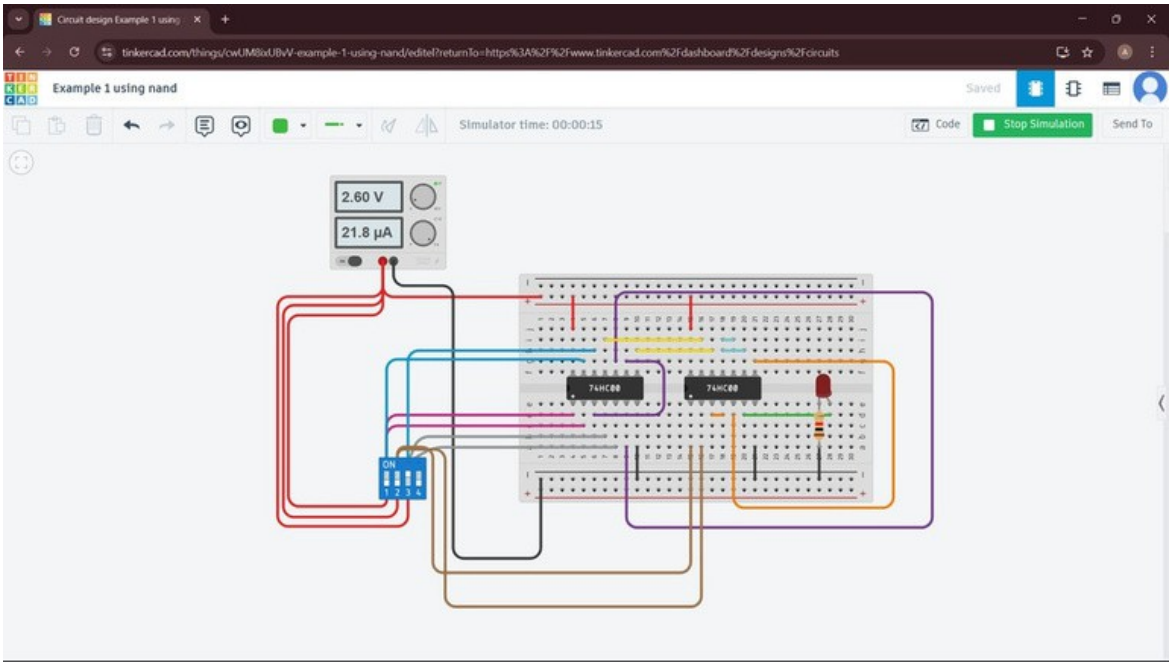
2) simplification with a Karnaugh map:

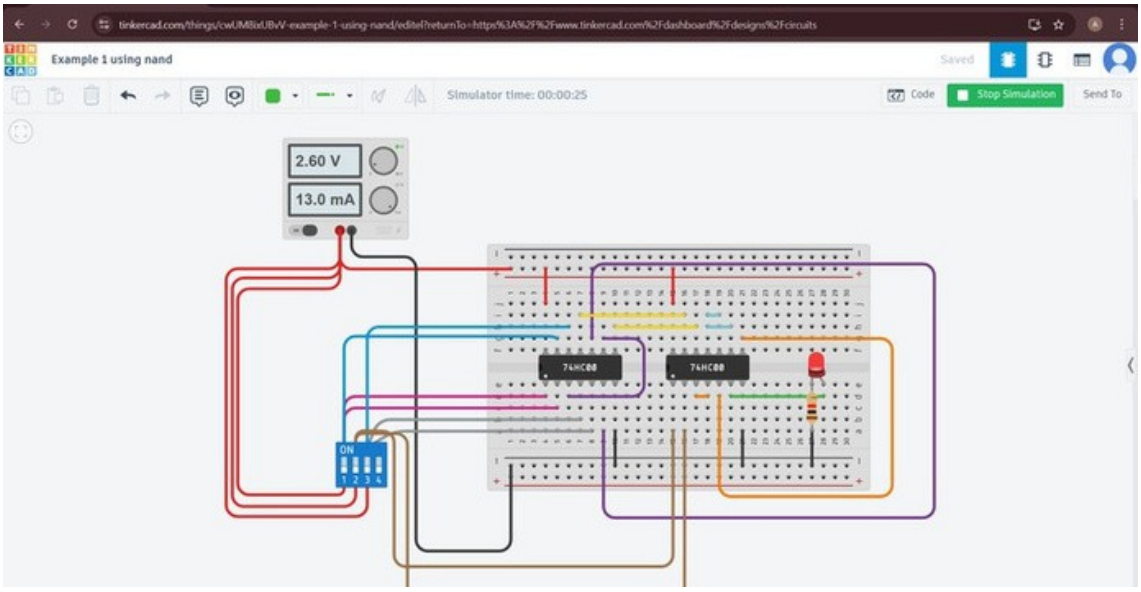


1) Using Nand gate:

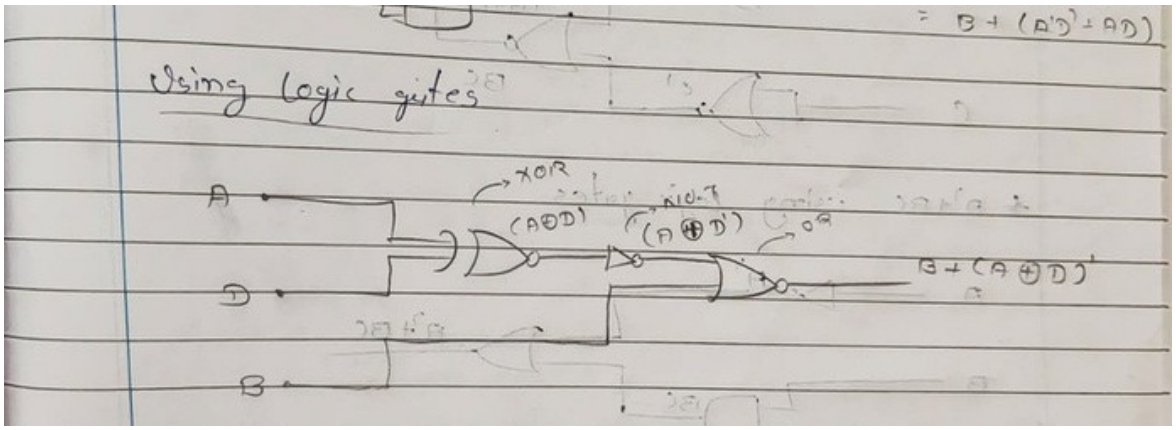


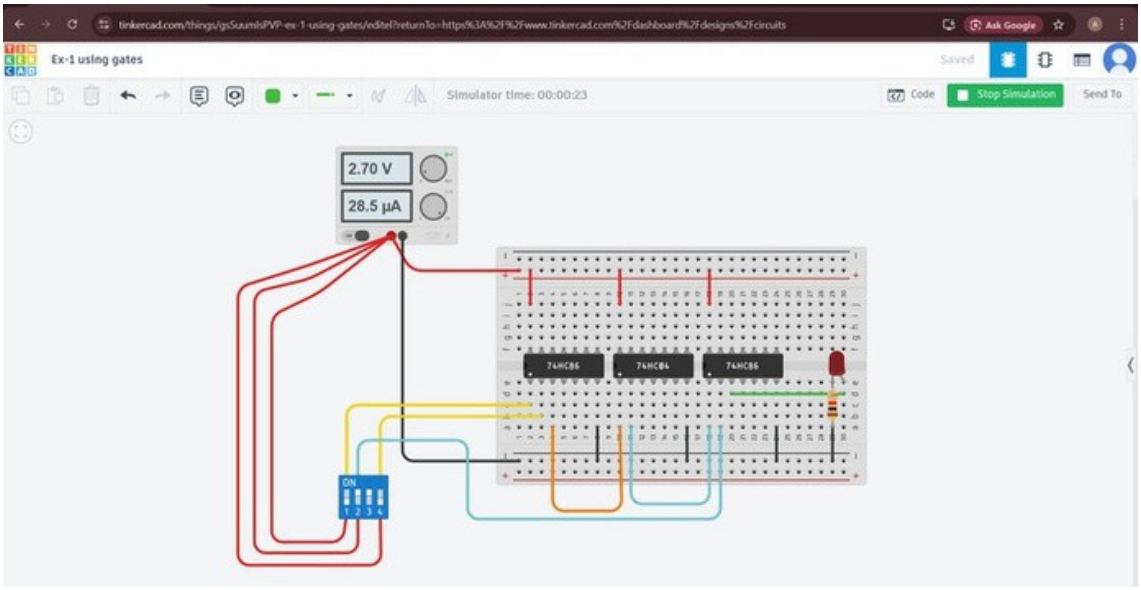
Output:

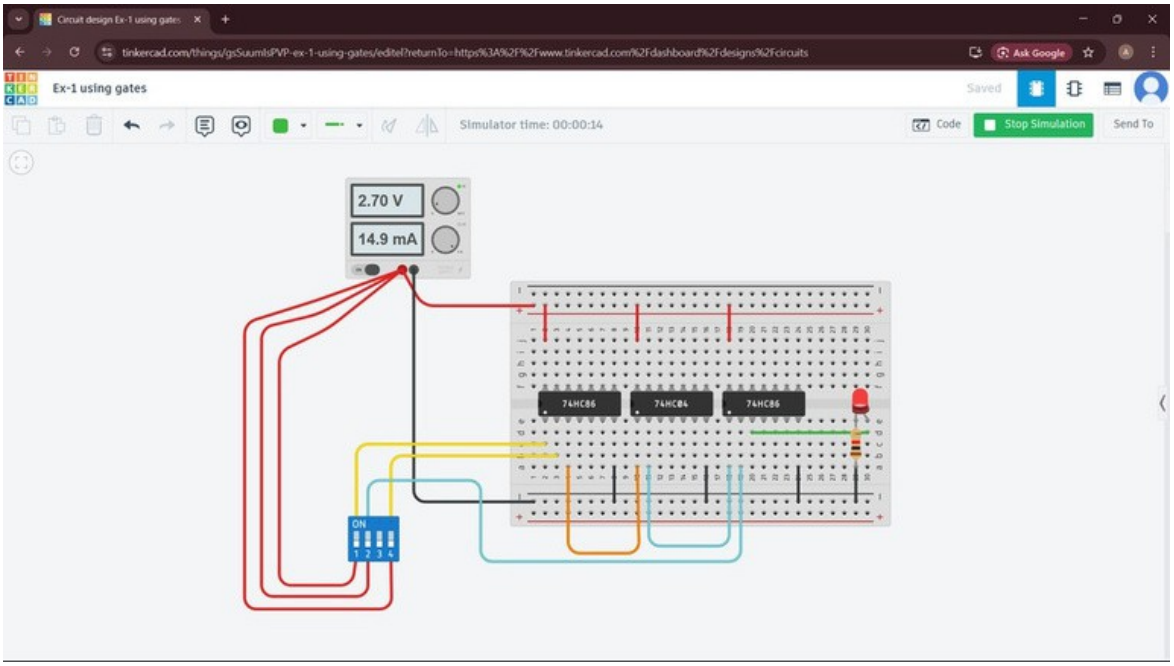




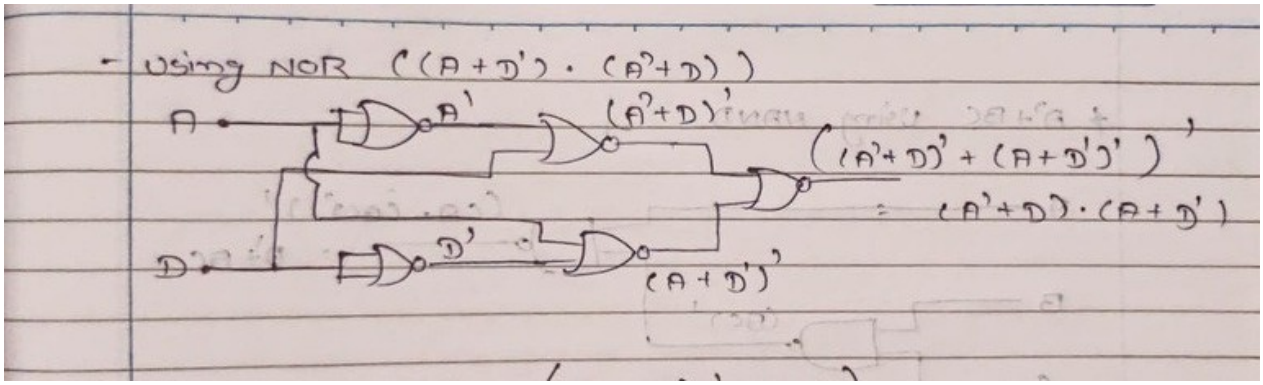
2) Using Logic gates:

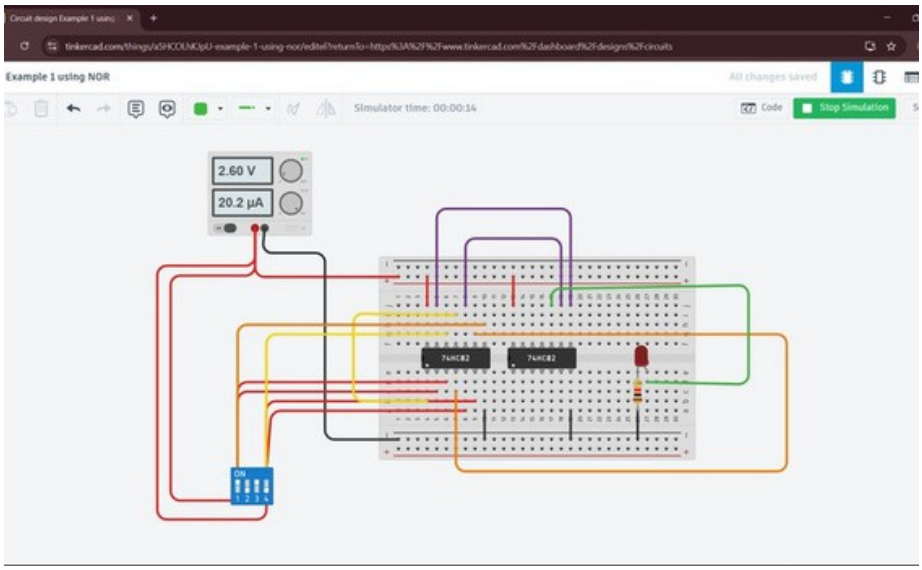



Ex-1 using gates
Simulator time: 00:00:23
Code
Stop Simulation
Send To
2.70 V
28.5 µA
74HC86
74HC04
74HC86
ON
1 2 3 4


Circuit design Ex-1 using gates
Ex-1 using gates
Simulator time: 00:00:14
Code
Stop Simulation
Send To
2.70 V
14.9 mA
74HC86
74HC04
74HC86
ON
1 2 3 4

3) Using Nor gate:





Output:

